Brook Mao

647-964-6882 | brook.mao@mail.utoronto.ca | linkedin.com/in/brook-mao | github.com/BrookMaoDev

Education

University of Toronto — Toronto, Canada
Honours Bachelor of Science in Computer Science — *Expected Apr 2028*

Experience

Software Engineer Intern — May 2025 – Aug 2026
Microchip Technology Inc. — *Toronto, Canada*

- Developed a new IDE for programming Microchip FPGAs on a team of 2–5 engineers
- Extended the open-source Theia IDE with custom logic and features using object-oriented TypeScript
- Identified issues with the Theia IDE and contributed fixes to the upstream open-source project
- Added AI features to the IDE, enabling another team to integrate their specialized coding assistant
- Fixed complex bugs in the IDE to improve reliability for users, and in CI pipelines to unblock team members
- Used Claude Code to accelerate development while making sure the team understood every change
- Enhanced the UI/UX using React and CSS based on Figma designs and customer feedback
- Added remote SSH support, allowing users to connect to remote servers and program attached hardware
- Completed and maintained a hardware register viewer with read/write support, extending it to newer FPGAs
- Upgraded dependencies and refactored extensive portions of the codebase to maintain compatibility
- Improved robustness of the IDE's Clangd C/C++ extension, providing users with more accurate error highlighting
- Wrote automated E2E tests with Playwright to catch regressions and reduce manual testing, saving developer time
- Completed a handed-off project to improve the readability of log output from projects using multiple processors
- Used Git, Docker, Make, Python, Bash, and Jenkins as part of the team's Linux workflow
- Developed a new PowerShell workflow for local development on Windows, giving team members more OS flexibility
- Reviewed dozens of pull requests, catching bugs early and pushing for better design decisions
- Authored documentation and trained customers on using the product during new releases
- Used Atlassian products such as Jira for issue tracking and Confluence for project planning and progress reports

Projects

YouTube Auto Comment | *TypeScript, Node.js, Express, PostgreSQL, GCP, Docker*

- Built a web app that monitors selected YouTube channels and posts a comment as soon as a new video drops
- Integrated Google OAuth 2.0 so the app can comment on behalf of users, with JWT cookies for sessions
- Designed a Postgres schema linking users, channels, and comments, using foreign keys to keep data consistent
- Deployed on Google Cloud Run with Docker, using Cloud Scheduler to poll YouTube for new uploads

ThreatTrace AI | *Python, Flask, TensorFlow, JavaScript, GCP*

- Collaborated in a team of 4 at Hack the Valley to develop a machine learning model that classifies cell images as healthy or malaria-infected with 95.45% accuracy, winning the Pulsar ML Challenge
- Coordinated tasks and clarified roles for teammates to deliver a working prototype within 36 hours
- Built the neural network using TensorFlow and trained it on a dataset of over 27,000 images, documenting the process in a Jupyter notebook
- Preprocessed the training data and user-uploaded images to ensure model compatibility
- Deployed the model via a web app using HTML, CSS, Bootstrap, and JavaScript to create a sleek frontend, and Python with Flask to develop a REST API backend, allowing users to try the model with their own images
- Set up a CI/CD pipeline with GitHub Actions, automating Docker image builds and deployments to GCP

MultiClock | *JavaScript, MERN, Tailwind CSS, NGINX, Docker, Linux, AWS, GitHub Actions*

- Developed a multiplayer chess clock web app using React, Node.js, Express, and MongoDB
- Designed a responsive UI to ensure the app works seamlessly on both desktop and mobile devices
- Deployed the app on AWS using a Linux VM, configuring NGINX as a reverse proxy with HTTPS encryption for secure public access
- Set up a CI/CD pipeline with GitHub Actions, automating Docker image builds and deployments to AWS

Concurrent System Monitor | *C, Linux, Make*

- Built a Linux CLI tool that reports hardware specs and tracks CPU and memory usage in real time
- Collected each metric in its own child process, piping results to the parent and handling signals to pause or quit

Linux File Descriptor Tables | *C, Linux, Make*

- Recreated Linux's per-process, system-wide, and vnode file descriptor tables by walking the /proc filesystem
- Resolved file paths and inode numbers for each open file, adding CLI flags to filter by PID and set thresholds

Mars Weather Data Pipeline | *Go, Google Cloud Storage, Google BigQuery, Power BI*

- Developed an ETL data pipeline in Go that extracts Mars weather data from the NASA API, processes it, and loads it into Google Cloud Storage and BigQuery
- Integrated Power BI with the BigQuery data warehouse to visualize Mars weather data and generate reports
- Built a Go API endpoint that automatically triggers the data pipeline

Tetris | *MIPS Assembly*

- Built Tetris in MIPS assembly with a partner, using memory-mapped I/O for the bitmap display and keyboard
- Implemented all 7 tetrominoes, collision detection, line clearing, scoring, high scores, and pause/game-over screens

Toronto Asian Art Museum Mobile App | *Android Studio, Java, Firebase*

- Developed an Android app using Java and Firebase Realtime Database for managing museum artifacts
- Led a team of 6 as Scrum master, facilitating meetings and sprint planning, achieving a 10/10 peer evaluation score
- Resolved bugs for 3 team members, ensuring on-time delivery of requirements within a 1-month deadline
- Implemented an interactive UI by extending AndroidX fragments to create custom elements
- Applied object-oriented design patterns like Singleton and Adapter to write maintainable code

Portfoliify | *PHP, MySQL, Apache, JavaScript, Azure*

- Developed a web app for creating and managing online profiles using PHP, MySQL, and JavaScript
- Designed a relational data model featuring 5 SQL tables to efficiently store user data
- Set up a CI/CD pipeline with GitHub Actions, automating Docker image builds and deployments to Azure

3D Rendering Engine | *C++, Cppcheck, Make*

- Built an object-oriented graphics program in C++ to render 3D shapes like cubes and pyramids
- Added zoom, pan, and rotation features using linear algebra
- Applied software design principles to ensure maintainability and facilitate future changes
- Used Makefiles to streamline the build process and conducted static code analysis with Cppcheck

Technical Skills

Languages: JavaScript, TypeScript, Python, C/C++, Java, SQL, Go, PHP, Bash, PowerShell, MIPS Assembly
Frameworks: React, Node.js, Express, Flask, JUnit, Jest, Playwright, Bootstrap, Tailwind CSS
Developer Tools: Git, Docker, Make, GitHub Actions, Jenkins, AWS, GCP, Azure, NGINX, Firebase, MongoDB, Linux
Libraries: pandas, NumPy, Matplotlib, TensorFlow